

BRISTOL, United Kingdom

WMO: 037243

Lat: 51.383N

Lon: 2.719W

Elev: 190

StdP: 99.07

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.2	-2.0	-6.0	2.3	-1.5	-4.8	2.6	-0.8	14.7	8.4	13.3	7.8	4.6	60	0.647

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	24.8	17.8	22.8	17.0	20.9	16.3	18.7	22.6	17.7	21.1	17.0	19.8	4.4	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	12.6	19.5	16.8	12.3	19.0	16.0	11.6	18.1	54.0	22.6	51.1	21.5	48.7	19.3	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.9	10.5	9.4	-5.1	27.7	1.8	2.3	-6.4	29.3	-7.4	30.7	-8.4	32.0	-9.7	33.6		
			-5.5	20.6	1.8	1.3	-6.8	21.5	-7.8	22.3	-8.8	23.0	-10.1	24.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.0	4.7	4.8	6.4	8.6	11.3	14.4	16.3	15.9	13.9	11.0	7.6	5.3
	DBStd	4.98	3.17	3.00	2.80	2.71	2.67	2.43	2.34	2.08	2.12	2.52	2.89	3.44
	HDD10.0	746	166	145	114	57	15	1	0	0	1	18	81	149
	HDD18.3	3047	424	378	370	292	217	122	74	81	134	228	323	405
	CDD10.0	759	1	1	2	15	57	132	194	184	117	48	8	2
	CDD18.3	18	0	0	0	0	0	3	9	6	1	0	0	0
	CDH23.3	121	0	0	0	0	2	26	63	28	2	1	0	0
	CDH26.7	14	0	0	0	0	0	3	7	4	0	0	0	0
Wind	WSAvg	4.9	5.5	5.4	5.3	4.9	4.7	4.5	4.4	4.2	4.4	4.9	4.9	5.3
Precipitation	PrecAvg	944	97	64	74	59	67	68	68	78	88	90	95	102
	PrecMax	1167	178	148	182	112	180	163	164	171	210	198	205	219
	PrecMin	658	18	2	4	4	8	11	14	19	20	7	43	21
	PrecStd	126	46	40	39	28	42	39	38	42	47	52	44	49
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.1	12.9	15.2	20.2	22.9	26.8	27.8	26.8	23.1	18.9	14.9	13.0
		MCWB	10.9	10.2	10.6	13.5	16.2	17.6	18.6	18.5	17.4	16.0	14.0	11.7
	2%	DB	11.0	10.9	12.9	16.9	20.1	23.1	25.1	23.1	20.2	16.2	13.2	12.0
		MCWB	10.1	9.5	9.8	12.0	15.0	16.7	17.8	18.1	16.2	14.6	12.3	11.2
	5%	DB	10.0	9.9	11.2	14.2	18.0	21.0	22.9	21.1	18.8	15.8	12.8	11.1
		MCWB	9.5	9.0	9.1	10.4	13.8	15.9	17.1	16.9	15.3	14.2	11.9	10.2
	10%	DB	9.0	9.0	10.2	12.9	16.0	19.0	20.9	19.8	17.2	14.8	11.9	10.1
		MCWB	8.7	8.3	8.7	9.7	12.8	14.7	16.4	16.3	14.7	13.4	10.9	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.3	11.0	11.6	14.1	17.5	19.5	19.7	19.8	18.5	16.8	14.2	12.2
		MCDB	11.9	12.3	13.8	18.5	20.8	24.3	25.0	23.5	21.4	18.3	14.7	12.6
	2%	WB	10.2	9.9	10.7	12.3	15.7	17.4	18.6	18.6	17.1	15.4	13.0	11.5
		MCDB	10.7	10.5	12.2	16.0	19.0	21.6	23.3	21.5	19.3	16.4	13.5	12.0
	5%	WB	9.2	9.0	9.9	11.2	14.4	16.3	17.6	17.6	16.2	14.5	12.0	10.6
		MCDB	9.5	9.4	10.8	13.7	17.2	19.9	21.5	20.2	18.0	15.5	12.6	11.1
	10%	WB	8.5	8.2	9.1	10.2	13.2	15.3	16.7	16.8	15.4	13.7	11.1	9.6
		MCDB	8.8	8.7	9.9	12.0	15.6	18.2	19.8	19.0	17.0	14.7	11.8	9.9
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.7	5.7	6.8	7.1	7.3	7.2	6.6	6.4	5.2	4.4	4.1
		MCDBR	4.9	5.1	7.0	9.3	10.0	10.9	11.0	8.9	8.3	5.6	4.5	4.4
		MCWBR	5.1	4.3	5.1	5.7	6.0	6.0	5.3	4.8	5.0	4.2	4.2	4.6
	5% WB	MCDBR	4.7	4.7	6.0	8.0	8.8	9.3	9.1	7.6	6.9	5.1	4.2	4.4
		MCWBR	5.0	4.6	4.8	5.4	5.6	5.6	5.1	4.6	5.0	4.4	4.3	4.7
Clear-Sky Solar Irradiance	taub	0.321	0.335	0.366	0.384	0.389	0.382	0.379	0.367	0.358	0.345	0.320	0.310	
	taud	2.375	2.400	2.313	2.275	2.298	2.344	2.356	2.408	2.429	2.444	2.423	2.351	
	Ebn at Noon	690	789	826	852	865	873	869	862	829	766	687	635	
	Edh at Noon	63	79	103	120	123	119	115	104	91	75	59	55	
All-Sky Solar Radiation	RadAvg	0.79	1.49	2.63	4.06	4.98	5.41	5.07	4.20	3.17	1.79	0.95	0.61	
	RadStd	0.07	0.18	0.34	0.45	0.25	0.57	0.47	0.36	0.23	0.15	0.09	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page